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Few Product Gates but Many Zeros

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Abstract A *d-gem* is a $\{+, -, \times\}$ -circuit having very few $\times$ -gates and computing from $\{x\} \cup \mathbb{Z}$ a univariate polynomial of degree d having d distinct integer roots. We introduce *d-gems* because they could help factoring integers and because their existence for infinitely many d would blatantly disprove a variant of the Blum-Cucker-Shub-Smale conjecture. A natural step towards validating the conjecture would thus be to rule out *d-gems* for large d . Here we construct *d-gems* for several values of d up to 55. Our 2^n -*gems* for $n \leq 4$ are skew, that is, each $\{+, -\}$ -gate adds an integer. We prove that skew 2^n -*gems* if they exist require n $\{+, -\}$ -gates, and that these for $n \geq 5$ would imply new solutions to the Prouhet-Tarry-Escott problem in number theory. By contrast, skew *d-gems* over the real numbers are shown to exist for every d .

1 Introduction

Consider the polynomials of degrees 4, 8 and 16 computed by the $\{+, \times\}$ -circuits depicted in Figure 1. Each polynomial factors completely over $\mathbb{Z}$ and its roots are distinct. Clearly the number of $\times$ -gates used in each case is minimum.

Main question: Do $\{+, -, \times\}$ -circuits having n $\times$ -gates and computing a polynomial $f(x) \in \mathbb{Z}[x]$ having 2^n distinct integer roots exist for all n ?

Crandall [Cr96, Prob. 3.1.13] found a *normalized* circuit for the case $n = 3$ and asked whether such circuits exist for $n > 3$, where a *normalized* circuit starts from x and alternates between a squaring operation and the addition of a constant. Dilcher [Di00] characterized the normalized circuits for $n = 3$. Crandall and Pomerance [CrPo01] constructed an example for $n = 4$ and Bremner [Br08] constructed two infinite families of examples for that case.

Why care? First, polynomials with distinct integer roots may hold the key to factoring integers. For example, knowing [BoMo74] that $\sim n^{1/4}$ operations modulo n suffice to evaluate $x(x-1)(x-2)\cdots(x-n^{1/4}+1)$ at the points $x = n^{1/4}, 2n^{1/4}, 3n^{1/4}, \dots, n^{1/2}$, Strassen [St76] noted that $\log_2 n$ -bit integers can be factored in time $\sim n^{1/4}$. No provably faster deterministic algorithm for factoring is known [GaGe03]. Lipton [Li94] later formulated a hypothesis, on circuits computing polynomials having many distinct integer roots, whose validity would imply that the integer factoring problem is “too easy” to support

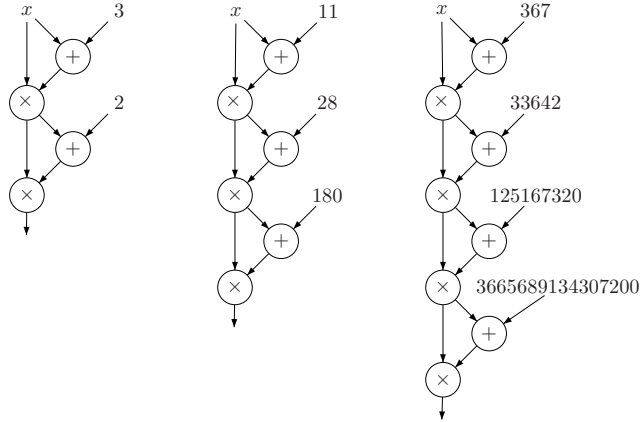


Figure 1. Circuits computing polynomials having the respective sets of roots $\{0, -1, -2, -3\}$, $\{0, -1, -2, -4, -7, -9, -10, -11\}$ and $\{0, -4, -7, -12, -118, -133, -145, -178, -189, -222, -234, -249, -355, -360, -363, -367\}$.

cryptography. A positive answer to our main question (with further size and constructivity assumptions) would validate Lipton’s hypothesis. Crandall [Cr96] discusses further connections with factoring.

Second, by Proposition 2.2, a positive answer would *refute* the L -conjecture of Bürgisser [Bu01]. The L -conjecture states that for some β and any d , any polynomial $f(x) \in \mathbb{Q}[x]$ has at most $(L(f) + d)^\beta$ irreducible factors of degree d or less, where $L(f)$ is the size of a smallest $\{+, -, \times, \div\}$ -circuit computing $f(x)$ from $\{x\} \cup \mathbb{Q}$ (see also [Ch04] and [Ro03]). The L -conjecture implies the τ -conjecture [BCSS97], namely that any $f(x) \in \mathbb{Z}[x]$ has at most $(\tau(f) + 1)^\beta$ distinct roots in $\mathbb{Z}$, where $\tau(f)$ is the size of a smallest $\{+, -, \times\}$ -circuit computing $f(x)$ from $x \cup \{1\}$. The τ -conjecture implies $P_{\mathbb{C}} \neq NP_{\mathbb{C}}$ in the Blum-Shub-Smale model of computation over the reals [BCSS97], and Smale named the τ -conjecture the fourth most important millennium mathematical challenge [Sm00]. A compelling micro-step towards validating both the τ -conjecture and the L -conjecture would thus be to provide a negative answer to our main question.

Our main question remains open. Here we partly explain why, by relating the question to a classical number-theoretic problem. We also extend the question to encompass polynomials of any degree $d > 0$ by defining d -gems, so named to reflect their “precious and seemingly rare” nature. Let ℓ_d be the length of a shortest addition chain for d , or equivalently, the size of a smallest $\{+\}$ -circuit (Section 2 defines circuits formally) computing d from $\{1\}$.

Definition 1.1. A $\{+, -, \times\}$ -circuit c with inputs from $\{x\} \cup \mathbb{Z}$ is a d -gem if c has at most ℓ_d $\times$ -gates and if the polynomial $f_c(x) \in \mathbb{Z}[x]$ computed by c has degree d and has precisely d distinct integer roots.

Hence a 2^n -gem is a $\{+, -, \times\}$ -circuit with n product gates that computes a polynomial of maximum degree 2^n and this polynomial factors completely with its 2^n roots integer and distinct. Our contributions are the following:

- we observe, following [PaSt73], that any polynomial of degree d can be computed from $\{x\} \cup \mathbb{Z}$ by a $\{+, -, \times\}$ -circuit using $2(\sqrt{d} + 1)$ product gates;
- if d -gems exist for infinitely many d , then the L -conjecture fails;
- by theoretical considerations and computer search, we construct *skew d -gems* for $d \leq 22$ and d -gems for $d \leq 31$ and $d = 36, 37, 42, 54, 55$ (see Figure 2), where a circuit is *skew* if each of its $\{+, -\}$ -gates merely adds an integer;
- whereas d -gems trivially require ℓ_d product gates when $d = 2^n$, we show that this holds for every $d \leq 71$; we conclude that all the d -gems we are able to construct so far have a minimal number of product gates;
- we observe that a skew 2^n -gem for any $n \geq 5$ would provide new solutions of size 2^{n-1} to the Prouhet-Tarry-Escott problem which has an almost 200-year history in number theory (see for instance [BoIn94]);
- we spell out sufficient conditions implying a 2^n -gem;
- we construct skew d -gems *over the reals*, i.e. with inputs from $\mathbb{R} \cup \{x\}$ and with the requirement of distinct roots in $\mathbb{R}$, for every d ;
- we prove that any skew 2^n -gem *over the reals* requires at least n $\{+, -\}$ -gates; we conclude that the skew 2^n -gems (over $\mathbb{Z}$) depicted in Figure 1 have a minimal number of $\{+, -\}$ -gates among all skew 2^n -gems.

Section 2 defines circuits and proves basic facts. Section 3 relates the existence of d -gems to the Prouhet-Tarry-Escott problem. Section 4 deals with gems over the real numbers. Section 5 describes our d -gem constructions. Section 6 concludes. Proofs left out of this abstract will appear in the full version of the paper.

2 Preliminaries and basic facts

By an (*arithmetic*) $\{+, -, \times\}$ -circuit c we mean a rooted directed acyclic graph with in-degree-2 nodes called *product gates* labeled with $\times$, in-degree-2 nodes called *additive gates* labeled with $+$ or $-$ and in-degree-0 nodes called *input gates* labeled with an integer or the variable x . We write $c_\times$ and c_+ for the numbers of product and additive gates in c respectively. The *size* of c is $c_\times + c_+$. A circuit c *represents* or *computes* a polynomial $f_c(x) \in \mathbb{Z}[x]$. A *zero* or *root* of c is an integer a such that $f_c(a) = 0$. We write $\mathbf{zeros}_c$ for the set of zeros of c . For example, if c is the leftmost circuit in Figure 1, then $c_\times = c_+ = 2$ and c represents $f_c(x) = (x(x+3))(x(x+3)+2) = x^4 + 6x^3 + 11x^2 + 6x$ having $\mathbf{zeros}_c = \{0, -1, -2, -3\}$.

An *addition chain* for a natural number d is an increasing sequence $d_0 = 1, d_1, \dots, d_k = d$ of natural numbers such that each d_i for $i > 0$ is the sum of two earlier numbers in the sequence. The polynomial x^d is computable by an optimal $\{\times\}$ -circuit having ℓ_d product gates, where ℓ_d is the minimum k for which there is an addition chain such that $d_k = d$ (see [Kn81] for extensive related facts on addition chains, such as $\lceil \log_2 d \rceil \leq \ell_d \leq 2\lceil \log_2 d \rceil$ for all d). Recall that ℓ_d enters Definition 1.1 of a d -gem c in the form “ $c_\times \leq \ell_d$ ”. We impose “ $c_\times \leq \ell_d$ ” rather than “ $c_\times$ is minimal” because of the prospect that Strassen (unpublished work) might have uncovered circuits c with $c_\times < \ell_{\deg(f_c(x))}$ and

d	$c_{\times}$	c_{+}	$f_c(x)$	izeros_c
1	0	0	x	$\{0\}$
2	1	1	$x^2 - 1$	$\{-1, 1\}$
3	2	1	$(x^2 - 1)x$	$\{-1, 0, 1\}$
4	2	2	$((x^2 - 5)^2 - 16)$	$\{-1, 1, -3, 3\}$
5	3	2	$((x^2 - 5)^2 - 16)x$	$\{0, -2, 2, -3, 3\}$
5	3	2	$(x^2 - 1)((x^2 - 4)x)$	$\{0, -1, 1, -2, 2\}$
6	3	2	$((x^2 - 25)^2 - 24^2)(x^2 - 25)$	$\{-1, 1, -7, 7, -5, 5\}$
6	3	2	$((x^2 - 7)x)^2 - 36$	$\{1, 2, -3, -1, -2, 3\}$
7	4	2	$((x^2 - 7)x)^2 - 36)x$	$\{0, -1, 1, -2, 2, -3, 3\}$
7	4	2	$((x^2 - 25)^2 - 24^2)(x^2 - 25)x$	$\{0, -1, 1, -7, 7, -5, 5\}$
8	3	3	$((x^2 - 65)^2 - 1696)^2 - 2073600$	$\{-3, 3, -11, 11, -7, 7, -9, 9\}$
9	4	2	$((x^2 - 49)x)^2 - 120^2)((x^2 - 49)x)$	$\{0, -3, 3, -5, 5, -8, 8, -7, 7\}$
10	4	3	$((y^2 - 236448)^2 - 123552^2)y$ with $y = (x^2 - 625)$	$\{\pm 5, 35, 17, 31, 25\}$
10	4	3	$((x^2 - 250)^2 - 14436)x^2 - 1612802$	$\{\pm 4, 8, 14, 18, 20\}$
12	4	3	$((x^2 - 91)x)^2 - 58500^2 - 50400^2$	$\{\pm 1, 5, 6, 9, 10, 11\}$
14	5	3	$((x^2 - 7^4)x)^2 - \dots)^2 - \dots)(x^2 - 7^4)$ $y \times (y^2 - 34320^2) \times (y^2 - 41160^2)$ with $y = (x^2 - 7^4)x$	$\{\pm 49, 16, 39, 55, 21, 35, 56\}$
15	5	3	$((x^2 - 7^4)x)^2 - 34320^2 \times (y^2 - 41160^2)$ with $y = (x^2 - 7^4)x$	$\{\pm 0, 49, 16, 39, 55, 21, 35, 56\}$
16	4	4	$((x^2 - 67405)^2 - 3525798096)^2 - \dots)^2 - \dots$	$\{\pm 11, 367, 131, 343, 77, 359, 101, 353\}$
18	5	5	$f_{c_{16}} \cdot (x^2 - 1)$	$\text{Set}_{16} \cup \{-1, 1\}$
18	5	4	$(y^2 - 2484^2) \times (y^2 - 4116^2) \times (y^2 - 5916^2)$ with $y = (x^2 - 7^2 \cdot 13)x$	$\{\pm 4, 23, 27, 7, 21, 27, 12, 17, 29\}$
20	5	5	$f_{c_{16}} \cdot ((x^2 - 67405)^2 - 3958423056)$	$\{\pm 67, 361, 11, 367, 131, 343, 77, 359, 101, 353\}$
21	6	4	$y \times (y^2 - 89760^2) \times (y^2 - 150480^2) \times$ $(y^2 - 263640^2)$ with $y = (x^2 - 7^2 \cdot 13^2)x$	$\{\pm 0, 91, 11, 85, 96, 19, 80, 99, 39, 65, 104\}$
22	6	6	$f_{c_{20}} \cdot (x^2 - 1)$	$\text{Set}_{20} \cup \{-1, 1\}$
23	6		$y \times (y - 4838400x^3 + 208051200x) \times (x^3 - 16x)$ with $y = z \times (z + 45x^3 - 700x^2 - 2835x + 630)$ and $z = (x^3 + x^2 - 197x + 195) \times (x^2 - x - 42)$	$\{\pm 0, 1, 2, 3, 4, 6, 7, 9, 10, 13, 14, 15\}$
24	5		$f_{c_4}(y^2)$ with $y = (x^2 - 7 \cdot 13 \cdot 19)x$	$\{\pm 3, 40, 43, 8, 37, 45, 15, 32, 47, 23, 25, 48\}$
24	5	442	$z(z + c_{Prop.3.4})$ with $y = (x^2 - 11763)^2$ and $z = (y + 241x^2 + ..)(y + 195x^2 + ..)(y + x^2 + ..)$	$\{\pm 22, 61, 86, 127, 140, 151, 35, 47, 94, 121, 146, 148\}$
26	6	443	$f_{c_{24}} \cdot (x^2 - 1)$	$\text{Set}_{24} \cup \{-1, 1\}$
27	6		$y \times f_{c_4}(y^2)$ with $y = (x^2 - 7^2 \cdot 13^2)x$	$\text{Set}_{21} \cup \{\pm 49, 56, 105\}$
28	6	560	$f_{c_{24}} \cdot (y + 117x^2 + \dots)$	$\text{Set}_{24} \cup \{-1, 1, -153, 153\}$
30	6		$f_{c_5}(y^2)$ with $y = (x^2 - 7^2 \cdot 13^2 \cdot 19)x$	$\{\pm 13, 390, 403, 35, 378, 413, 70, 357, 427, 103, 335, 438, 117, 325, 442\}$
36	6		$f_{c_6}(y^2)$ with $y = (x^2 - 7^2 \cdot 13^2 \cdot 19)x$	$\text{Set}_{30} \cup \{\pm 137, 310, 447\}$
42	7		$f_{c_7}(y^2)$ with $y = (x^2 - 7^2 \cdot 13^2 \cdot 19)x$	$\text{Set}_{36} \cup \{\pm 182, 273, 455\}$
54	7		$f_{c_9}(y^2)$ with $y = (x^2 - 7^2 \cdot 13^2 \cdot 19)x$	$\text{Set}_{42} \cup \{\pm 202, 255, 457, 225, 233, 458\}$
55	8		The above $\times x$	The above $\cup \{0\}$

Figure 2. Some d -gems which are skew for $d \leq 22$. When two examples are given for a given d , these arise from different minimal addition chains for d . The functions f_{c_i} are from Lemma 5.1 if $i \leq 9$ and from the i -gem in this table otherwise. We omitted the cases $d = 11, 13, 17, 19, 25, 29, 31, 37$ which, like the case $d = 55$ here, are obtained by extending a $(d - 1)$ -gem; we note that such an extension does not work for 43 which has a shorter addition chain bypassing 42. The constructions not mentioned is this abstract are explained in the full paper.

we do not want declaring the gem status to require a lower bound proof. When $d = 2^n$, then $2^{\ell_{2^n}} = 2^n = d = |\text{izers}_c| = \deg(f_c(x)) \leq 2^{c_\times} \leq 2^{\ell_d} = 2^{\ell_{2^n}}$ so $c_\times$ is indeed then minimal. We extend this to all d -gems constructed in this paper as follows, since $d < 71$ implies $d > 2^{d-3}$ (see [Kn81, p. 446]):

Lemma 2.1. *If $d > 2^{d-3}$, then any $\{\times, +, -\}$ -circuit computing a polynomial $f(x) \in \mathbb{Z}[x]$ of degree d requires at least ℓ_d product gates.*

The L -conjecture clearly fails if skew d -gems exist for infinitely many d , but this also holds for d -gems:

Proposition 2.2. *If d -gems exist for infinitely many values of d , then the L -conjecture fails.*

To qualify as a d -gem, a circuit c must be extremely efficient in terms of its number of product gates, that is, $c_\times \leq \ell_d \leq 2 \log_2 d$. At the opposite end of the spectrum, we have:

Proposition 2.3. [PaSt73] *Any degree- d polynomial $f(x) \in \mathbb{Z}[x]$ can be computed by a $\{+, -, \times\}$ -circuit having at most $2\sqrt{d} + 1$ product gates.*

Proof. Write $f(x) = (\cdots ((g_k x^k + g_{k-1})x^k + g_{k-2})x^k + \cdots + g_1)x^k + g_0$ where each $g_i \in \mathbb{Z}[x]$ has degree $k = \lceil \sqrt{d} \rceil$. Once $x^2, x^3, \dots, x^k$ are available, each g_i is computable using additive gates alone. Another k products by x^k suffice. $\square$

3 Gems and the Prouhet-Tarry-Escott problem

For any $n > 4$, we are unable to rule out the existence of a skew 2^n -gem. In this section we show that any such gem would yield new solutions to a number-theoretic problem having a long history. Then we examine whether solutions to the number-theoretic problem can help constructing gems in return.

Recall from Section 1 that a 2^n -gem is *normalized* if its computation starts from x and iterates the sequence “squaring then adding a constant” at least once. A normalized 2^n -gem thus has n $\times$ -gates and n $+$ -gates and is entirely described by a sequence $\gamma_1, \dots, \gamma_n$ of integers with $n \geq 1$. Normalized 8-gems and normalized 16-gems are investigated in [Cr96, Di00, CrPo01, Br08].

Consider an arbitrary skew 2^n -gem c . Consecutive $+$ -gates in c can be merged, i.e. $(g + a) + b$ for a gate g and $a, b \in \mathbb{Z}$ can be rewritten $g + (a + b)$. And for c to reach degree 2^n , each $\times$ -gate g must reach twice the degree of the nearest $\times$ -gate g' having a path to g , i.e. g must perform $(g' + a) \times (g' + b)$ with $a, b \in \mathbb{Z}$. Hence c can be taken to be the circuit $\text{Skew}(\alpha_0, \beta_0, \dots, \alpha_{n-1}, \beta_{n-1}, \alpha_n)$ depicted in Figure 3.

Proposition 3.1. *Let $\alpha_0, \dots, \alpha_n, \beta_0, \dots, \beta_{n-1}, a_1, \dots, a_{2^n} \in \mathbb{Z}$ be such that $c = \text{Skew}(\alpha_0, \beta_0, \dots, \alpha_{n-1}, \beta_{n-1}, \alpha_n)$ computes the polynomial $p(x) = \prod_{i=1}^{2^n} (x - a_i)$. For any $t \in \mathbb{Z} \setminus \{0\}$, $c' = \text{Skew}(\alpha_0 t^{2^0}, \beta_0 t^{2^0}, \dots, \alpha_{n-1} t^{2^{n-1}}, \beta_{n-1} t^{2^{n-1}}, \alpha_n t^{2^n})$ computes the polynomial $q(x) = \prod_{i=1}^{2^n} (x - ta_i)$.*

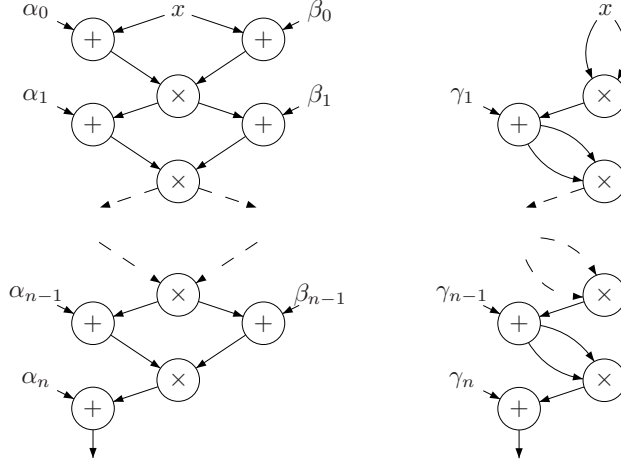


Figure 3. The skew 2^n -gem $Skew(\alpha_0, \beta_0, \dots, \alpha_{n-1}, \beta_{n-1}, \alpha_n)$ and its normal form.

Proof. We claim that c' computes $q(x) = t^{2^n} p(x/t)$. This concludes the proof since then $q(ta_1) = \dots = q(ta_{2^n}) = 0$. So let $n = 0$. Then c' computes $x + \alpha_0 t^{2^0} = t \cdot (x/t + \alpha_0) = t \cdot p(x/t)$. Now consider $n > 0$ and let q_1, q_2 and p_1, p_2 be computed by the gates input to the lowest $\times$ -gate in c' and c respectively. Then $q(x) = q_1(x) \times q_2(x) + \alpha_n t^{2^n} = t^{2^{n-1}} p_1(x/t) \times t^{2^{n-1}} p_2(x/t) + \alpha_n t^{2^n}$ by induction, and the latter equals $t^{2^n} [p_1(x/t) \times p_2(x/t) + \alpha_n] = t^{2^n} p(x/t)$. $\square$

Lemma 3.2. (Normal form) Let $n \geq 1$. Given $a_1, \dots, a_{2^n} \in \mathbb{Z}$ and a skew 2^n -gem c such that $f_c(x) = \prod_{i=1}^{2^n} (x - a_i)$, there exist $s \in \mathbb{Z}$ and $t \in \{1, 2\}$ and a normalized 2^n -gem computing the polynomial $\prod_{i=1}^{2^n} (x - ta_i - s)$.

Proof. Let $c = Skew(\alpha_0, \beta_0, \dots, \alpha_{n-1}, \beta_{n-1}, \alpha_n)$ (see Figure 3). If $\alpha_i + \beta_i$ is odd for some $0 \leq i < n$ then let $t = 2$, else let $t = 1$. By Proposition 3.1, the skew 2^n -gem $c' = Skew(\alpha_0 t^{2^0}, \beta_0 t^{2^0}, \dots, \alpha_{n-1} t^{2^{n-1}}, \beta_{n-1} t^{2^{n-1}}, \alpha_n t^{2^n})$ computes $q(x) = \prod_{i=1}^{2^n} (x - ta_i)$. We will now normalize c' . First we rewrite each $\times$ -gate $(g + a) \times (g + b) = g^2 + (a + b)g + ab$ as $[g + (a + b)/2]^2 + [ab - ((a + b)/2)^2]$, noting that any such $a + b$ occurring in c' is even. Then we merge consecutive $+$ -gates since these are skew. The result would be normalized, were it not for an extraneous $x + s$ gate at the input level. We replace $x + s$ with x . This yields a normalized 2^n -gem computing $q(x - s) = \prod_{i=1}^{2^n} (x - ta_i - s)$. $\square$

Definition 3.3. (see [BoIn94]) Two sets $\{a_1, \dots, a_m\}, \{b_1, \dots, b_m\}$ solve the PTE (Prouhet-Tarry-Escott) problem of degree k if $a_1^i + \dots + a_m^i = b_1^i + \dots + b_m^i$ for all $i \leq k$. A solution is called ideal if $k = m - 1$. A solution of the form $\{a_1, -a_1, \dots, a_{m/2}, -a_{m/2}\}, \{b_1, -b_1, \dots, b_{m/2}, -b_{m/2}\}$ is called symmetric and we abbreviate it by $\{a_1, \dots, a_{m/2}\}, \{b_1, \dots, b_{m/2}\}$; a solution of the form $\{a_1, \dots, a_m\}, \{-a_1, \dots, -a_m\}$ is also called symmetric.

Let $p(x) = (x - a_1)(x - a_2) \cdots (x - a_m)$ and $q(x) = (x - b_1)(x - b_2) \cdots (x - b_m)$. Define, for $k = 1, 2, \dots, m$, $s_k = \sum_{i=1}^m a_i^k$ and $t_k = \sum_{i=1}^m b_i^k$.

Proposition 3.4. ([DoBr37], see [BoIn94].) *The following are equivalent:*

- $s_1 = t_1$ and $s_2 = t_2$ and $s_3 = t_3$ and $\cdots$ and $s_k = t_k$
- $\deg[p(x) - q(x)] \leq m - (k + 1)$.

Corollary 3.5. *For any $n \geq 1$ and any skew 2^n -gem c such that $\text{izeros}_c = \{a_1, \dots, a_{2^n}\}$, there is a partition $S \uplus T = \{a_1, \dots, a_{2^n}\}$ into two equal size sets such that the pair S, T is an ideal PTE solution of size 2^{n-1} .*

Proof. We apply Lemma 3.2 to c and obtain a circuit c' computing a polynomial $(r(x))^2 + \gamma_n = \prod_{i=1}^{2^n} (x - ta_i - s)$ for some $\gamma_n \in \mathbb{Z}$ (Figure 3), $s \in \mathbb{Z}, t \in \{1, 2\}$. Hence $-\gamma_n = e$ for some $e \in \mathbb{N}$. Now $(r(x))^2 - e = p(x)q(x)$, where $p(x) = r(x) + \sqrt{e}$ and $q(x) = r(x) - \sqrt{e}$. Since $\mathbb{Z}[x]$ is a Euclidian ring, $p(x)$ and $q(x)$ must each have 2^{n-1} distinct roots. (So $\sqrt{e} \in \mathbb{N}$.) Now $\deg(p(x) - q(x)) = 0$, so applying Proposition 3.4 with $k = 2^{n-1} - 1$ shows that $\{a \in \mathbb{Z} : p(a) = 0\}$ and $\{a \in \mathbb{Z} : q(a) = 0\}$ form an ideal PTE solution of size 2^{n-1} . It is well known [BoIn94] that shifting from $ta_i + s$ to ta_i preserves PTE solutions. Further dividing out the ta_i by t to get back to the a_i also preserves PTE solutions. $\square$

A skew 32-gem, if it exists, thus implies an ideal PTE solution of degree 15. Borwein and Ingalls [BoIn94, p. 8] state that “it has been conjectured for a long time that ideal PTE solutions exist for every n ”, yet the largest such solution known today is due to Shuwen [Shu01] and has degree 11 [We09].

For the converse direction, consider the ideal symmetric PTE solution $\{2, 16, 21, 25\}$, $\{5, 14, 23, 24\}$ of degree 7 [BoIn94, p. 8]. We could try to unravel the Corollary 3.5 construction, by expressing $p(x) = (x^2 - 2^2)(x^2 - 16^2)(x^2 - 21^2)(x^2 - 25^2)$ using 3 products. Here this happens to be possible, by calculating $(x^2 - 2^2)(x^2 - 25^2)$ using 2 products, then forming $(x^2 - 16^2)(x^2 - 21^2)$ from $(x^2 - 2^2)(x^2 - 25^2)$ by repeatedly subtracting x^2 (unavoidable since $2^2 + 25^2 \neq 16^2 + 21^2$), and finally obtaining $p(x)$ using one last product. In this special case, we could therefore construct a (non-skew) 16-gem from an ideal symmetric PTE solution.

Figure 2 contains further gems constructed with the help of PTE solutions, such as a 24-gem constructed with the help of a degree-11 solution. But even if PTE solutions of degree 15 were known, these would need to fulfill additional properties in order to yield 32-gems by the strategy described above. We now show that the “sym-perfect” condition below is a sufficient additional condition imposed on ideal symmetric PTE solutions to yield a normalized 2^n gem.

Definition 3.6. *A pair $S, T \subset \mathbb{Z}$ is called sym-perfect if $S = \{a\}$ and $T = \{-a\}$, or if $S = \{a_1, -a_1, \dots, a_{m/2}, -a_{m/2}\}$, $T = \{b_1, -b_1, \dots, b_{m/2}, -b_{m/2}\}$ and $\exists s \in \mathbb{Z}$ such that $\{a_1^2 + s, \dots, a_{m/2}^2 + s\}, \{b_1^2 + s, \dots, b_{m/2}^2 + s\}$ is sym-perfect.*

If a_1^2 and $a_{m/2}^2$ are respectively smallest and largest in $\{a_1^2, \dots, a_{m/2}^2\}$, then the shift s occurring in the recursive definition of sym-perfect is necessarily equal

to $-(a_1^2 + a_{m/2}^2)/2$ (which is also the average of all the numbers that occur). An example of a sym-perfect pair is $\{-3, 3, -11, 11\}, \{-7, 7, -9, 9\}$, since the pair $\{9 - 65, 121 - 65\}, \{49 - 65, 81 - 65\}$ is sym-perfect by virtue of the pair $\{56^2 - 1696\}, \{16^2 - 1696\}$ in turn being sym-perfect.

Theorem 3.7. *A set $U \subset \mathbb{Z}$ of size 2^{n+1} can be written as $S \cup T$ for a sym-perfect pair S, T if and only if U is the set of zeros of a normalized 2^{n+1} -gem.*

Proof: The sym-perfect pair $\{a\}, \{-a\}$ corresponds to the 2-gem c with $f_c(x) = x^2 - a^2$ and $\mathbf{zeros}_c = \{a, -a\}$, which forms the induction basis $n = 0$. Now let $U = \{a_1, -a_1, \dots, a_{2^n/2}, -a_{2^n/2}\} \cup \{b_1, -b_1, \dots, b_{2^n/2}, -b_{2^n/2}\}$ where the pair $S' = \{a_1^2 - s, \dots, a_{2^n/2}^2 - s\}$, $T' = \{b_1^2 - s, \dots, b_{2^n/2}^2 - s\}$ is sym-perfect. By induction, $S' \cup T'$ is $\mathbf{zeros}_{c'}$ for a 2^n -gem c' . Hence U is $\mathbf{zeros}_c$ for the 2^{n+1} -gem c with $f_c(x) = f_{c'}(x^2 - s)$. Conversely, a normalized 2^{n+1} -gem c with zero set U computes $f_c(x) = g(x)^2 + \gamma_{n+1} = f_{c_S}(x) \times f_{c_T}(x)$ for some normalized 2^n -gems c_S and c_T where γ_i is the same as in c for all $i < n$ (see Figure 3). Then $U = S \cup T$ for their zeros $S = \{a_1, -a_1, \dots, a_{2^n/2}, -a_{2^n/2}\}$ and $T = \{b_1, -b_1, \dots, b_{2^n/2}, -b_{2^n/2}\}$. The pair S', T' as above with $s = -\gamma_1$ for γ_1 from the gem c is sym-perfect by induction, which makes S, T sym-perfect. $\square$

Corollary 3.8. *A sym-perfect pair is an ideal, symmetric PTE solution.*

Proof: We have to show that $a_1^i + (-a_1)^i + \dots + a_{m/2}^i + (-a_{m/2})^i = b_1^i + (-b_1)^i + \dots + b_{m/2}^i + (-b_{m/2})^i$ for all $i < m$. For odd i , neighbours cancel. For $i = 2i'$, this is double the i' -th equation of the PTE $\{a_1^2, \dots, a_{m/2}^2\}, \{b_1^2, \dots, b_{m/2}^2\}$ which is just a shift of the sym-perfect PTE from the induction. $\square$

4 The case of real numbers

Exclusively in this section, we extend Definition 1.1 to the case of real numbers, that is, to $\{+, -, \times\}$ -circuits having inputs from $\{x\} \cup \mathbb{R}$ and computing polynomials that factor completely over $\mathbb{R}$. We will bound c_+ from below for skew 2^n -gems c over $\mathbb{R}$ and then construct skew d -gems over $\mathbb{R}$ for every d . Let A be $\mathbb{Z}, \mathbb{Q}$ or $\mathbb{R}$. For short, we will say that a nonzero polynomial $p(x) \in \mathbb{R}[x]$ *crumbles over A* if $\deg(p) = 0$ or if p has $\deg(p)$ distinct roots in A .

Proposition 4.1. *Let A be $\mathbb{Z}, \mathbb{Q}$ or $\mathbb{R}$. Let $p(x) \in \mathbb{R}[x]$ and $q(x) \in \mathbb{R}[x]$. If pq crumbles over A then both p and q crumble over A .*

Fact 4.2. (Rolle) *Over $\mathbb{R}$, the derivative of a crumbling polynomial crumbles.*

Note that Rolle only applies over $\mathbb{R}$. For example, $(x-1)(x-2)(x-3)$ crumbles over $\mathbb{Z}$ but its derivative $3x^2 - 12x + 11$ crumbles neither over $\mathbb{Z}$ nor over $\mathbb{Q}$. For that reason, the only proof we have of Corollary 4.5 below is via Theorem 4.4.

Lemma 4.3. *Let $e \in \mathbb{R}$ and suppose that a polynomial $p(x) + e \in \mathbb{R}[x]$ of degree 2^n crumbles over $\mathbb{R}$. Then the following holds:*

- H1.** *Any skew gem over $\mathbb{R}$ for $p(x)$ has at least $n - 1$ additive gates.*
- H2.** *If $e = 0$ then any skew gem over $\mathbb{R}$ for $p(x)$ has at least n additive gates.*

Theorem 4.4. *A skew 2^n -gem over the reals has at least n additive gates.*

Proof. This follows by applying Lemma 4.3 with $e = 0$. $\square$

Corollary 4.5. *Any skew 2^n -gem (over $\mathbb{Z}$) requires n additive gates.*

Proof. Let a skew 2^n -gem c compute $p(x) \in \mathbb{Z}[x]$. If c had fewer than n additive gates, then c as a skew gem over the reals would contradict Theorem 4.4. $\square$

Rojas [Ro03, p. 4] constructs 2^n -gems over $\mathbb{R}$ for any n . The following variation constructs skew 2^n -gems: $g_1 = x$ and $g_{i+1}(x) := g_i^2(x) - 2$, $1 \leq i < n$, yields $g_n(x)$ having 2^n distinct roots in $[-2, 2]$. We extend this to arbitrary degrees:

Proposition 4.6. *For all $d > 0$, there exists a skew d -gem over $\mathbb{R}$.*

The construction in the proof of Proposition 4.6 produces at most ℓ_d additions (exactly ℓ_d when $d = 2^n$). In cases like $d = 3, 7, 9, 27, 81$, it produces $\ell_d/2$ additions. We conjecture that $c_+ \geq \ell_d/2$ for a skew d -gem c over $\mathbb{R}$.

5 Constructing gems

In this section we construct gems (over $\mathbb{Z}$), at times with the help of a computer. Yet we do not know whether 32-gems or d -gems beyond $d = 55$ exist.

Lemma 5.1. *For every $d \leq 7$ and $d = 9$, for every d distinct integers $a_1, \dots, a_d$, there is a d -gem c_d such that $f_{c_d}(x) = (x - a_1)(x - a_2) \cdots (x - a_d)$.*

Proof. The d -gems are: $f_{c_1}(x) = (x - a_1)$, $f_{c_2}(x) = (x - a_1) \times (x - a_2)$,
 $f_{c_3}(x) = f_{c_2}(x) \times (x - a_3)$,
 $f_{c_4}(x) = f_{c_2}(x) \times (f_{c_2}(x) + \underbrace{(a_1 + a_2 - a_3 - a_4) \cdot x + (a_3 a_4 - a_1 a_2)}_{\text{iterated additions of } x})$,
 $f_{c_5}(x) = f_{c_4}(x) \times (x - a_5)$,
 $f_{c_6}(x) = f_{c_3}(x) \times (f_{c_3}(x) + a \cdot f_{c_2}(x) + (a \cdot (a_1 + a_2) - a_1 a_2 - a_1 a_3 - a_2 a_3) \cdot x + (a_4 a_5 a_6 - a_1 a_2 a_3 - a \cdot a_1 a_2))$ with $a = (a_1 + a_2 + a_3 - a_4 - a_5 - a_6)$,
 $f_{c_7}(x) = f_{c_6}(x) \times (x - a_7)$,
 $f_{c_9}(x) = f_{c_6}(x) \times (f_{c_3}(x) + a \cdot f_{c_2}(x) + (a \cdot (a_1 + a_2) - a_1 a_2 - a_1 a_3 - a_2 a_3) \cdot x + (a_7 a_8 a_9 - a_1 a_2 a_3 - a \cdot a_1 a_2))$ with $a = (a_1 + a_2 + a_3 - a_7 - a_8 - a_9)$. $\square$

The number of additive gates used in proving Lemma 5.1 can be reduced when the roots satisfy favorable conditions, such as $a_1 = -a_2$ or $a_i = 0$ for some i . Such relationships between the zeros are exploited extensively in Figure 2.

The case $d = 8$ is missing from Lemma 5.1 because not all polynomials having 8 distinct roots have an 8-gem [Di00]. For any $0 < a < b < c < d$, we can easily construct an 8-gem for $(x+a)(x-a)(x+b)(x-b)(x+c)(x-c)(x+d)(x-d)$ by prepending the 4-gem (available by Lemma 5.1) for $(y-a^2)(y-b^2)(y-c^2)(y-b^2)$ by “ $y \leftarrow x \times x$ ”.

To construct a 16-gem, we seek $0 < a < b < c < d < e < f < g < h$ and an 8-gem for $(y-a^2)(y-b^2)(y-c^2)(y-d^2)(y-e^2)(y-f^2)(y-g^2)(y-h^2)$. We first develop sufficient conditions for the existence of a skew 2^n -gem for any n .

Definition 5.2. (*Litter conditions.*) Let T be the full ordered binary tree with 2^n leaves labeled $a_1, a_2, \dots, a_{2^n} \in \mathbb{Z}$ in the natural order. Let each internal node in T be labeled with the product of the labels of the leaves subtended by that node. The sequence $a_1, a_2, \dots, a_{2^n}$ satisfies the litter conditions if, for $1 < i < n$, each of the 2^i nodes at level i has the same litter sum, where the litter sum of a node is defined as the sum of the labels of its two children.

Example 5.3. The litter conditions are mute for sequences of length 1 or 2. The litter conditions for the sequence a_1, a_2, a_3, a_4 are the equation $a_1 + a_2 = a_3 + a_4$. The litter conditions for the sequence $a^2, b^2, c^2, d^2, e^2, f^2, g^2, h^2$ are

$$a^2 + b^2 = c^2 + d^2 = e^2 + f^2 = g^2 + h^2 \quad \text{and} \quad a^2 b^2 + c^2 d^2 = e^2 f^2 + g^2 h^2. \quad (1)$$

Lemma 5.4. If the sequence $a_1, a_2, \dots, a_{2^n} \in \mathbb{Z}$ satisfies the litter conditions, then the following skew 2^n -gem c computes $p(x) = \prod_{1 \leq i \leq 2^n} (x - a_i)$:

$$\begin{aligned} y_0 &\leftarrow x - a_1 \\ y_1 &\leftarrow y_0 \times (y_0 + (a_1 - a_2)) \\ y_2 &\leftarrow y_1 \times (y_1 + (a_3 a_4 - a_1 a_2)) \\ y_3 &\leftarrow y_2 \times (y_2 + (a_5 a_6 a_7 a_8 - a_1 a_2 a_3 a_4)) \\ &\vdots \\ y_n &\leftarrow y_{n-1} \times (y_{n-1} + (\prod_{i=2^{n-1}+1}^{2^n} a_i - \prod_{i=1}^{2^{n-1}} a_i)). \quad \square \end{aligned}$$

We return to our quest for a 16-gem. By Lemma 5.4, any distinct squares $a^2, b^2, c^2, d^2, e^2, f^2, g^2, h^2$ satisfying (1) from Example 5.3 are the zeros of an 8-gem $p(y)$. In turn, each such 8-gem prepended by “ $y \leftarrow x \times x$ ” is a 16-gem. A small computer in a few hours found several examples, such as:

Proposition 5.5. A 16-gem with 4 additive gates exists to compute the polynomial having the 16 roots $\{\pm 237, \pm 106, \pm 189, \pm 178, \pm 227, \pm 126, \pm 218, \pm 141\}$.

We note that Bremner [Br08] focusses on 16-gems and constructs two infinite families. Turning to 32-gems, we were unable to find a 16-gem having 16 distinct squares as zeros, nor to find a 32-gem by appealing to the litter conditions of a sequence of length 32 directly.

The next lemma is our tool to generate d -gems when $d \neq 2^n$:

Lemma 5.6. Let $h(x) \in \mathbb{Z}[x]$ and $m_1, m_2, \dots, m_d \in \mathbb{Z}$. Suppose that each one of the d polynomials $h(x) - m_i$ is computed by a gem and that no two such polynomials share a root. If $\ell_d + \ell_{\deg(h)} \leq \ell_{d \cdot \deg(h)}$ and for some d -gem c , $f_c(y) = (y - m_1)(y - m_2) \cdots (y - m_d)$, then there is a gem computing $f_c(h(x))$.

We illustrate the use of Lemma 5.6 in the following:

Theorem 5.7. There exist 36-gems and 54-gems.

We have filled the rest of Figure 2 largely by trying to combine the mentioned methods along possible shortest addition chains for the degree d . Many cases remain open.

6 Conclusion

The following heuristic for factoring a $2n$ -bit integer $N = pq$, for primes p and q of comparable size, is inspired by Lipton [Li94]:

- assume distinct $a_i \in \mathbb{Z}$ and a circuit c computing $f_c(x) = \prod_{i=1}^{2^n} (x - a_i) \in \mathbb{Z}[x]$
- pick $a \in \{0, \dots, N - 1\}$ at random
- compute $d = f_c(a)$ modulo N by evaluating each gate in c modulo N
- output $\gcd(d, N)$.

This is merely a heuristic because its success probability depends on the distribution of the 2^n integers $(a - a_i)$ modulo N . If this is close to uniform, then indeed $\text{Prob}[1 < \gcd(d, N) < N]$ is constant. Of course the heuristic runs in time polynomial in the number of bits required to represent c , and the τ -conjecture [BCSS97] claims that this number is exponential in n .

Here we introduced *gems*. These are circuits that use an almost optimal number of $\times$ -gates to compute polynomials that factor completely over $\mathbb{Z}$ with distinct roots. A 2^n -gem could thus serve in the heuristic above if its inputs modulo N can be computed in time polynomial in $n = O(\log N)$, but the L -conjecture [Bu01] claims that even circuits that are much less constrained than d -gems do not exist for large d .

We exhibited d -gems over $\mathbb{R}$ for every d . But the d -gems we care about (over $\mathbb{Z}$) are elusive. In particular, constructing *skew* 32-gems would yield new solutions to the Prouhet-Tarry-Escott problem. These new PTE solutions would even fulfill additional conditions. Yet skew 2^n -gems for any $n \geq 5$ cannot currently be ruled out. This attests to the difficulty of the L -conjecture [Bu01], since skew 2^n -gems would provide the most severe counter-examples imaginable to it.

We constructed d -gems for several d up to $d = 55$. We proved that any skew 2^n -gem requires n additive gates. We showed that for $d \leq 71$, no circuit can compute a degree- d polynomial using fewer than ℓ_d product gates. Hence all the gems constructed so far are $\times$ -optimal, and our skew 2^n -gems for $n \leq 4$ are also $\{+, -\}$ -optimal among skew gems.

Our main open question is essentially the challenge we started with: do d -gems exist for infinitely many d ? A concrete step would be to undertake a search for d -gems of every type for small values of d , extending the systematic approach used by Bremner to study normalized 16-gems [Br08]. This would add entries to Figure 2. Hopefully it could help finding skew 2^n -gems for $n = 5$. As seen above, resolving the existence question for skew 2^n -gems seems like a natural baby step towards resolving the L -conjecture. But even this step would seem to break ground from a number-theoretic perspective.

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